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Battery Cell with Venting Cover and Battery Module Including the Same

Abstract

A battery cell includes: a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; a vent provided in the case or the cap plate; and a venting cover disposed to cover the vent, and provided to be opened by pressure acting from the accommodating space to the outside of the case, wherein the venting cover includes a venting sheet provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent and a second surface opposite to the first surface; and a venting cap having a support portion supporting the first surface of the venting sheet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0030275 filed on Feb. 29, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a battery cell having a venting cover and a battery module including the same.

BACKGROUND

[0003] A secondary battery, unlike a primary battery, has convenience, in that it can be charged and discharged, and is attracting a lot of attention as a power source for various mobile devices and electric vehicles.

[0004] Such a secondary battery may include a battery cell in which an electrode assembly formed by stacking a cathode plate, an anode plate, and a separator or by winding the same in a roll shape is accommodated inside a case together with an electrolyte. A plurality of battery cells may be stacked in a predetermined direction and accommodated in a battery module or battery pack.

[0005] Some types of secondary batteries may include a vent connecting the inside of the case and the outside of the case for communication therebetween to allow gas generated inside the case to be released to the outside of the case when an event such as thermal runaway has occurred. However, there may be a problem that flames or gas flowing from the exterior of the case through the vent can also flow into the interior of the case.

[0006] The above description is provided to help understand the technical background of the present disclosure, and should not be construed as reducing, limiting, or restricting the technical idea of the present disclosure. In addition, the contents described or suggested in the above description do not necessarily refer to prior art, and some may include contents that do not fall under prior art.

SUMMARY

[0007] An aspect of the present disclosure may provide a battery cell that can discharge flames, gas, or the like, within an accommodating space to the outside of a case when an event has occurred, but can prevent flames, gas, or the like from the outside of the case, from flowing into the accommodating space.

[0008] An aspect of the present disclosure may provide a battery module that can discharge flames, gas, or the like, within the accommodating space to the outside of a case when an event has occurred, but can prevent flames or gas, generated within an adjacent battery cell from flowing thereinto.

[0009] In addition, a battery cell and a battery module including the same of the present disclosure may be widely applied in green technology fields such as electric vehicles, battery charging stations, and solar power generation and wind power generation using batteries. In addition, the battery cell and battery module of the present disclosure may be used in eco-friendly electric vehicles, hybrid vehicles, or the like, to prevent climate change by suppressing air pollution and greenhouse gas emissions.

[0010] According to an aspect of the present disclosure, a battery cell may include: a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; a vent provided in the case or the cap plate; and a venting cover disposed to cover the vent, and provided to be opened by pressure acting from the accommodating space to the outside of the case, wherein the venting cover includes a venting sheet provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent and a second

surface opposite to the first surface; and a venting cap having a support portion supporting the first surface of the venting sheet.

[0011] In addition, according to an embodiment, the venting sheet may include a venting line formed on at least one of the first surface or the second surface.

[0012] In addition, according to an embodiment, the venting sheet may be disposed to face the vent in a first direction, and the support portion may be disposed to correspond to the venting line, so that at least a portion of the support portion may be provided to overlap the venting line in the first direction.

[0013] In addition, according to an embodiment, the venting line may be formed by being partially recessed from the second surface.

[0014] In addition, according to an embodiment, the vent may be provided to be opened by a predetermined amount of pressure and provided to connect the accommodating space and the exterior of the case for communication therebetween.

[0015] In addition, according to an embodiment, the vent may include a notch groove formed by being at least partially recessed in a thickness direction.

[0016] In addition, according to an embodiment, the venting cover may be provided to be opened when pressure within the accommodating space becomes greater than pressure outside the case by a predetermined range, to be opened by force toward the outside of the case from the accommodating space.

[0017] In addition, according to an embodiment, the venting cover further includes a venting space formed between the venting sheet and the vent, the vent may be provided to be opened by a pressure difference between the accommodating space and the venting space, and the venting sheet may be provided to be opened toward the outside of the case when the vent is opened.

[0018] In addition, according to an embodiment, the vent includes a notch recessed at least partially in the first direction.

[0019] In addition, according to an embodiment, the support portion is disposed between the venting sheet and the vent.

[0020] In addition, according to an embodiment, further comprising a terminal portion disposed on the cap plate and electrically connected with the electrode assembly, wherein the terminal portion includes: a cathode terminal electrically connected with a cathode tab of the electrode assembly; and an anode terminal electrically connected with an anode tab of the electrode assembly.

[0021] In addition, according to an embodiment, the venting cover is disposed between the cathode terminal and the anode terminal.

[0022] In addition, according to an embodiment, the vent is disposed between the cathode terminal and the anode terminal.

[0023] In addition, according to an embodiment the venting lines is plurality, and the plurality of venting lines are arranged in a second direction perpendicular to the first direction.

[0024] In addition, according to an embodiment, the support portion is provided in plurality, and the plurality of support portions are arranged in the second direction.

[0025] According to an aspect of the present disclosure, a battery cell may include: a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; and a venting cover provided in the cap plate or the case, and provided to be opened in a direction toward the outside of the case from the accommodating space, wherein the venting cover includes: a venting sheet provided to be opened by a predetermined amount of pressure, and including a venting line formed on at least one of a first surface facing the accommodating space and a second surface opposite to the first surface, in a first direction; and a venting cap having a support portion supporting the first surface of the venting sheet, wherein at least a portion of the support portion is provided to overlap the venting line in the first direction.

[0026] According to an aspect of the present disclosure, a battery module may include: a plurality of battery cells; and a module housing accommodating the plurality of battery cells, wherein at

least one of the plurality of battery cells includes a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; a vent provided in the case or the cap plate; and a venting cover disposed to cover the vent, and provided to be opened by a pressure acting from the accommodating space to the exterior of the case, wherein the venting cover includes: a venting sheet provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent and a second surface opposite to the first surface; and a venting cap having a support portion supporting the first surface of the venting sheet.

[0027] As described above, the solution according to the present disclosure has been described, but this is exemplary, and it should be understood that other configurations that are not mentioned are also included in the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0028] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0029] FIG. 1 is a perspective view of a battery cell according to an embodiment of the present disclosure.

[0030] FIG. 2 is a schematic internal cross-sectional view taken along line I-I' of FIG. 1.

[0031] FIG. 3 is an exploded view of a venting cap.

[0032] FIG. 4 is an enlarged view of A in FIG. 2.

[0033] FIG. 5 is a diagram exemplarily illustrating that a venting cover is opened to the outside of the case.

[0034] FIG. 6 is a diagram exemplarily illustrating that reverse inflow of gas flowing from the outside of the case is prevented.

[0035] FIG. 7 is a battery module according to an embodiment of the present disclosure.

[0036] FIG. 8 is a diagram exemplarily illustrating that an event has occurred in one battery cell of the battery module.

DETAILED DESCRIPTION

[0037] Prior to the detailed description of the present disclosure, the terms or words used in the present specification and claims described below should not be construed as being limited to a common or dictionary meaning, and the inventor intends to use his/her invention in the best way. Based on the principle that terms may be properly defined for description, they should be interpreted as meanings and concepts consistent with the technical spirit of the present disclosure.

[0038] The same reference numbers or symbols described in each drawing represent parts or components that perform substantially the same functions. For convenience of description and understanding, the same reference numbers or symbols may be used in different embodiments.

[0039] In the following description, singular expressions include plural expressions unless the context clearly dictates otherwise. It should be understood that terms such as “include” or “comprise” are intended to specify the presence of a feature, number, step, operation, component, part, or combination thereof described in the specification, but do not exclude in advance the possibility of the presence or addition of one or more other features, numbers, steps, operations, components, parts, or combinations thereof.

[0040] In addition, in the following description, expressions such as upper side, upper portion, lower side, lower portion, side surface, front, rear, etc. are expressed based on the direction illustrated in the drawings, and may be expressed differently if the direction of the object is changed.

[0041] In addition, terms including ordinal numbers such as “first,” “second,” or the like. may be

used to distinguish between components in the description and claims below. These ordinal numbers are used to distinguish between identical or similar components, and the meaning of the terms should not be limited due to the use of these ordinal numbers. For example, the components associated with these ordinal numbers should not be interpreted in a restricted order of use or disposition by their numbers. If necessary, respective ordinal numbers may be used interchangeably.

[0042] Hereinafter, the present disclosure will be described in detail with reference to the drawings.

[0043] FIG. 1 is a perspective view of a battery cell according to an embodiment of the present disclosure, and FIG. 2 is a schematic internal cross-sectional view taken along line I-I' of FIG. 1.

[0044] Referring to FIGS. 1 and 2, a battery cell **10** of the present disclosure may include a case **110** accommodating an electrode assembly **120** in an accommodating space S, a cap plate **130** coupled to at least one side of the case **110**, a vent **133** provided in the case **110** or the cap plate **130**, and a venting cover **200** disposed to cover the vent **133** but provided to be opened by pressure acting from the accommodating space S to the outside of the case **110**. Here, the venting cover **200** may include a venting sheet **230** provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent **133** and a second surface opposite to the first surface, and a venting cap **210** having a support portion supporting the first surface of the venting sheet **230**.

[0045] The case **110** may provide an accommodating space S in which an electrode assembly **120** to be described later can be accommodated. In the drawing, the case **110** is expressed as a roughly rectangular shape, but this is only an example, and the case may be changed into various shapes as long as it has a structure in which an accommodating space S for receiving the electrode assembly **120** is formed.

[0046] In addition, a cap plate **130**, to be described later, may be coupled to an opening (not shown) of the case **110** to seal the accommodating space S.

[0047] A material of the case **110** may be appropriately selected in consideration of thermal and electrical conductivity, rigidity corresponding to swelling of the electrode assembly **120**, processability, manufacturing cost, or the like. For example, the case **110** may be formed of a metal material including aluminum, aluminum alloy, or the like.

[0048] The electrode assembly **120** may be disposed in the accommodating space S of the case **110**. The electrode assembly **120** may include a cathode material **121**. The cathode material **121** may include a cathode current collector and a cathode active material. In some embodiments, the cathode current collector may include aluminum, an aluminum alloy, or the like, and the cathode active material may include lithium cobalt oxide, lithium manganese oxide, lithium nickel oxide, lithium iron phosphate, or the like. The cathode active material may be coated on a surface of the cathode current collector. Portions of the cathode current collector, uncoated with the cathode active material, may function as cathode tabs **120a**. In some embodiments, the cathode tab **120a** may be provided in plural, and some or all of the plurality of cathode tabs **120a** may be bonded to each other.

[0049] The electrode assembly **120** may include an anode material **122**. The anode material **122** may include an anode current collector and an anode active material. In some embodiments, the anode current collector may include copper, a copper alloy, nickel, a nickel alloy, or the like, and the anode active material may include carbon, silicon, or the like. The anode active material may be coated on a surface of the anode current collector. Portions of the anode current collector, uncoated with an anode active material, may function as an anode electrode tab **120b**. In some embodiments, the anode tab **120b** may be provided in plural, and some or all of the plurality of anode tabs **120b** may be bonded to each other.

[0050] The electrode assembly **120** may include a separator **123**. The separator **123** may be disposed between a cathode material **121** and an anode material **122**. The separator **123** may function to provide a passage for the movement of ions, while physically contacting the cathode material **121** and the anode material **122**. For example, the separator **123** may include a porous

polymer film or a porous substrate. In some embodiments, the separator **123** may be formed of a polymer material including polyethylene, polypropylene, or the like. In addition, the separator **123** may include a dry and wet separator. In some embodiments, the separator **123** may include a coating layer including a ceramic coating layer or the like.

[0051] The electrode assembly **120** may be formed by disposing the above-described components in a method such as winding, stacking, or the like. For example, the electrode assembly **120** may be formed in a structure in which the cathode material **121**, the anode material **122**, and the separator **123** are wound around an axis in a longitudinal direction or transverse direction. Alternatively, the electrode assembly **120** may be formed as a structure in which the above-described winding structure is compressed in a direction approximately perpendicular to a winding axis. The above-described winding structure may be referred to as a ‘jelly roll’ in the art.

[0052] As another example, the electrode assembly **120** may be formed in a structure in which a cathode material **121**, an anode material **122**, and a separator **123** are sequentially stacked in a vertical direction (e.g., in a thickness direction or height direction of the electrode assembly). In some cases, the separator **123** in the stacked structure may be formed as a stacked structure in which a plurality of unit separators **123**, continuous in the longitudinal direction, are sequentially folded according to the stack of the cathode material **121** and the anode material **122**. The above-described stacked structure may be referred to in the art as ‘stack and folding,’ ‘z-folding,’ or the like. However, in this embodiment, the disposition of each component of the electrode assembly **120** is not particularly limited. The electrode assembly **120** may have various dispositions other than those exemplified above.

[0053] In some embodiments, the electrode assembly **120** may be formed by combining a plurality of unit units. For example, the electrode assembly **120** may include unit units wound in a jelly roll manner, and two or more of the unit units may be combined to form an electrode assembly **120**. In the present embodiment, the electrode assembly **120** is comprised of two jelly roll units. As another example, the electrode assembly **120** may include a unit unit wound in a stack-and-fold manner, and two or more of the unit units may be combined to form an electrode assembly **120**.

[0054] The electrode assembly **120** may be accommodated in an accommodating space **S** of the case **110** together with an electrolyte. In some embodiments, the electrolyte can be formed of an organic solvent containing a lithium salt. For example, the lithium salt may include lithium hexafluorophosphate (LiPF₆), lithium tetrafluoroborate (LiBF₄), or the like, in a liquid or gel state, and the organic solvent may include cyclic carbonates such as ethylene carbonate (EC), propylene carbonate (PC), or the like, and linear carbonates such as diethyl carbonate (DEC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), or the like.

[0055] In some other embodiments, the electrolyte may be omitted or replaced. For example, when an inorganic solid electrolyte is used, the liquid or gel-type electrolyte may be omitted.

[0056] In addition, the structure of the electrode assembly **120** illustrated in the present disclosure is exemplary. In the present disclosure, an electrode assembly **120** including a cathode tab **120a** and an anode tab **120b** arranged in parallel from one side of the electrode assembly **120** is illustrated, but the positions and/or numbers of the cathode tab **120a** and the anode tab **120b** are exemplary. For example, in an embodiment not shown, the cathode tab **120a** and the anode tab **120b** may be disposed with the electrode assembly **120** (e.g., cathode material **121**), anode material **122**, and separator **123**) interposed therebetween. At least a portion of the cathode tab **120a** and the anode tab **120b** may be disposed between the case **110** and the electrode assembly **120**, respectively.

[0057] The cap plate **130** may be coupled to the case **110** to close an opening of the case **110**. In the drawing, the cap plate **130** is illustrated as a square plate shape corresponding to the opening of the case **110**.

[0058] The cap plate **130** may be coupled to the case **110** to seal the accommodating space **S** of the case **110** in which the electrode assembly **120** is disposed. In some embodiments, the cap plate **130** may be welded to the case **110** by ultrasonic welding, laser welding, or the like. Terminal portions

131a and **131b** electrically connected to the electrode assembly **120** may be disposed on the cap plate **130**. The battery cell **10** may be electrically connected to an external power source through the terminal portions **131a** and **131b**. The terminal portions **131a** and **131b** may include a cathode terminal **131a** and an anode terminal **131b**. The cathode terminal **131a** may be electrically connected to the cathode tab **120a** of the electrode assembly **120**, and the anode terminal **131b** may be electrically connected to the anode tab **120b** of the electrode assembly **120**.

[0059] The cap plate **130** may include a vent **133**. In the present disclosure, it is illustrated that the vent **133** is disposed between the cathode terminal **131a** and the anode terminal **131b**, but the position of the vent **133** may be variously changed as needed and is not necessarily limited to what is exemplified. For example, in another embodiment not shown, a vent **133** be disposed or added to the case **110**.

[0060] The vent **133** may be formed to be opened in response to internal pressure of the case **110**. In other words, the vent **133** may be provided to be opened by a predetermined amount of pressure, thereby allowing the accommodating space(s) and the outside of the case **110** to be in communication. Thereby, the vent **133** can function to discharge the internal pressure to the outside of the case **110** and contribute to stabilizing internal elements of the case **110**.

[0061] According to an embodiment of the present disclosure, the vent **133** may include a notch **133a** of a predetermined shape. The notch **133a** may be a vulnerable portion formed to be at least partially recessed in the thickness direction of the vent **133**. Since a fracture of the vent **133** is induced through the notch by the difference between pressure within the accommodating space(S) or pressure outside the case **110**, the accommodating space(S) of the case **110** may be in communication with the outside of the case.

[0062] Meanwhile, in the present disclosure, a secondary battery **100** including a case **110** including one opening and a cap plate **130** disposed on one side of the case **110** is illustrated, but this is exemplary. In an embodiment not illustrated, a secondary battery **100** including a plurality of cap plates **130** disposed on both sides of the case **110** may be provided. For example, the secondary battery **100** may include a first cap plate disposed on one side of the case **110** and a second cap plate disposed on the other side of the case **110**.

[0063] In addition, the present disclosure may include a liquid injection unit **134** that allows fluid to flow by connecting the accommodating space(S) of the case **110** and the exterior of the case **110** for communication therebetween. The liquid injection unit **134** may be provided so that a fluid, such as an electrolyte, flowing into the accommodating space(S) can flow. The liquid injection unit **134** may include an injection hole (not shown) formed in the cap plate **130** and an injection cap (not shown) sealing the injection hole. The fluid may be injected into the accommodating space(S) through the injection hole and then the injection hole may be closed with the injection cap.

[0064] Hereinafter, a venting cover **200** of the present disclosure will be described in detail with reference to the drawings.

[0065] FIG. **3** is an exploded view of a venting cap, and FIG. **4** is an enlarged view of A in FIG. **2**.

[0066] Referring to FIGS. **3** and **4** together, the venting cover **200** may include a venting cap **210** including a support portion **213** and a venting sheet **230** mounted on the support portion **213** of the venting cap **210** and covering the vent **133**.

[0067] The venting cap **210** can support the venting sheet **230**. The venting cap **210** may be disposed on a location in which the vent **133** is provided among the cap plate **130** and the case **110** and support the venting sheet **230**. The venting cap **210** may have a hole formed, in communication with the vent **133** therein and include a support portion **213** crossing the hole. The venting cap **210** may support the venting sheet **230** through the support portion **213**.

[0068] The support portion **213** may be disposed to be spaced apart from the vent **133** by a predetermined distance in a first direction (Z-axis direction). Thereby, the venting cover **200** may further include a venting space (V) between the vent **133** and the venting sheet **230** therein.

[0069] The venting sheet **230** may be installed in the support portion **213** to block the hole of the

venting cap **210**. The venting sheet **230** may include a first surface (lower surface) facing the vent **133** and a second surface (upper surface) opposite thereto.

[0070] The venting sheet **230** may have a venting line **233**, at least partially recessed in a thickness direction, formed therein. The venting line **233** may be formed on at least one of the first surface (lower surface) and the second surface (upper surface). That is, the venting line **233** may be a vulnerable portion provided to be the first to be fractured in the venting sheet **230**.

[0071] The support portion **213** of the venting cap **210** may be provided to support the venting line **233**. In other words, the support portion **213** may be provided so that at least a portion thereof overlaps the venting line **233** in the first direction (Z-axis direction), to correspond to the venting line **233**. That is, the support portion **213** may be provided to support a vulnerable portion of the venting sheet **230**.

[0072] Meanwhile, the venting line **233** may be formed by being recessed from the second surface (upper surface) of the venting sheet **230**. Thereby, a contact area between the venting sheet **230** and the support portion **213** may be increased. However, as described above, this is merely an example of the present disclosure. For example, the venting line **233** may be provided by being recessed from the first surface (lower surface) of the venting sheet **230**. As another example, the venting line **233** may be formed on both the first surface (lower surface) and the second surface (upper surface) of the venting sheet **230**.

[0073] Meanwhile, according to another embodiment of the present disclosure, only a venting cover **200** may be provided without a vent **133**. According to another embodiment of the present disclosure, a case **110** accommodating the electrode assembly **120**, a cap plate **130** coupled to at least one side of the case **110**, and a venting cover **200** provided in the cap plate **130** or the case **110**, and provided to be opened in a direction toward the outside of the case **110** from the accommodating space S may be included. Here, the venting cover **200** may include a venting sheet **230** provided to be opened by a predetermined amount of pressure and including a venting line **233** formed on at least one of a first surface (lower surface) facing the accommodating space S in the first direction (Z-axis direction) or a second surface (upper surface) opposite to the first surface; and a venting cap **210** having a support portion **213** supporting the first surface (lower surface) of the venting sheet **230**. In this case, the support portion **213** may be provided so that a portion thereof overlaps the venting line **233** in the first direction (Z-axis direction), so that the venting sheet **230** is not fractured even when force is applied downwardly (−Z-axis direction). In the other embodiment, the configuration of the venting cover **200** overlaps with those described above, and is therefore omitted.

[0074] Meanwhile, in the present disclosure, a space inside the case **110** may be defined as an accommodating space S, a space between the vent **133** and the venting sheet **230** may be defined as a venting space V, and the outside of the case **110** may be referred to as an external space.

[0075] The vent **133** may be provided to be fractured and opened by pressure between the accommodating space S and the venting space V. The venting sheet **230** may be provided to be fractured by pressure between the venting space V and the external space.

[0076] In other words, the pressure generated within the accommodating space S is first applied to the vent **133**, and conversely, the pressure generated outside the case **110** may be first applied to the venting sheet **230**. Here, if the venting sheet **230** is not fractured by the pressure from gas, or the like, generated outside the case **110**, the vent **133** may not be opened. According to an embodiment of the present disclosure, the venting sheet **230** is provided to not be opened by the pressure generated outside the case **110**, thereby preventing gas, flames, or the like, flowing from the outside of the case from flowing into the interior of (accommodating space: S) the case **110**. That is, according to the present disclosure, the venting cover **200** may be selectively opened depending on the direction of the force applied to the venting cover **200**.

[0077] Hereinafter, a structure of a venting cover which is selectively opened depending on a direction in which pressure is applied is described with reference to the drawings.

[0078] FIG. 5 is a diagram exemplarily illustrating that a venting cover is opened to the outside of a case, and FIG. 6 is a diagram exemplarily illustrating that a reverse inflow of gas or the like flowing from the outside of the case is prevented.

[0079] Referring to FIG. 5, flames or gas (G) generated within an accommodating space S and discharged to the outside thereof ruptures the vent **133** and enters a venting space V, and then pressurizes the venting sheet **230** to bend upwardly (in a +Z-axis direction), thereby rupturing the venting sheet **230**. In this case, the venting sheet **230** may be fractured along a venting line **233**. Thereby, flames or gas (G) generated within the accommodating space (S) may be discharged the outside of the battery cell **10**.

[0080] As described above, the venting cover **200** may be provided to be opened when the pressure in the accommodating space (S) becomes greater than the pressure outside the case **110** by a predetermined range, to be opened by force toward the outside of the case **110** in the accommodating space.

[0081] In addition, in an embodiment of the present disclosure, the vent **133** may be provided to be opened by a pressure difference between the accommodating space (S) and the venting space (V), and the venting sheet **230** may be provided to be opened toward the outside of the case **110** when the vent **133** is opened.

[0082] On the other hand, referring to FIG. 6, flames or gas (E) flowing outside the battery cell **10** presses the venting sheet **230** downwardly (in a -Z axis direction) from the outside of the venting sheet **230**. In this case, the venting sheet **230** is supported by the support portion **213**, so the venting sheet **230** may not be bent in a pressing direction. In particular, since the support portion **213** supports a lower surface of a portion in which the venting line **233**, which is a vulnerable portion of the venting sheet **230**, is formed, the venting line **233** may be prevented from being fractured by external flames or gas (E). This can prevent the external flames or gas (E) from rupturing the vent **133** and flowing into the accommodating space (S). In addition, particles such as conductive particles included in the external flames or gas (E) may be prevented from flowing back into the accommodating space (S).

[0083] As described above, the battery cell **10** of the present disclosure may allow flames or gas (G) generated within the accommodating space (S) to flow toward the exterior, but prevent or minimize flames or gas (E) generated from the outside thereof from flowing into the accommodating space (S).

[0084] That is, whether the venting cover **200** is opened may be determined depending on the pressing direction (force) applied to the venting cover **200**.

[0085] FIG. 7 is a diagram illustrating a battery module according to an embodiment of the present disclosure, and FIG. 8 is a diagram exemplarily illustrating that an event has occurred in one battery cell in a battery module.

[0086] Referring to FIG. 7, a battery module **50** according to an embodiment of the present disclosure may accommodate a plurality of battery cells **10**. Here, the plurality of battery cells **10** may include at least one battery cell **10** of the present disclosure described above in FIGS. 1 to 6.

[0087] In other words, the battery module **50** according to an embodiment of the present disclosure may include a plurality of battery cells **10** and a module housing **510** accommodating the plurality of battery cells **10**. Here, at least one of the plurality of battery cells **10** may include a case **110** accommodating an electrode assembly **120** in an accommodating space S, a cap plate **130** coupled to at least one side of the case, a vent **133** provided in the case **110** or the cap plate **130**, and a venting cover **200** disposed to be opened by pressure acting from the accommodating space S to the exterior of the case. Here, the venting cover **200** may include a venting sheet **230** provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent **133** and a second surface opposite to the first surface, and a venting cap **210** having a support portion **213** supporting the first surface of the venting sheet **230**.

[0088] The housing **510** may include a lower cover **512** having an internal space (M) formed

therein and an upper cover **511** coupled to the lower cover **512** to close the internal space (M).

[0089] In addition, the battery module **50** of the present disclosure may include a plurality of bus bars **520** electrically connecting the plurality of battery cells **10** to each other. The bus bar **520** may be formed of an electrically conductive material such as metal and may electrically connect terminals **131a** and **131b** of adjacent battery cells **10** to each other. In an embodiment not shown, the housing **510** may further include a module terminal electrically connected to an external power source, and the module terminal may be electrically connected to the bus bar **520**.

[0090] Referring to FIG. **8**, if an event such as thermal runaway occurs in one (TS) of the plurality of battery cells **10**, flames or gas (E) generated within the battery cell **10** may be ejected into the internal space (M). The flames or gas (E) may increase the pressure within the internal space (M) and flow toward an adjacent battery cell (BS), and as described above, the venting cover **200** of the adjacent battery cell (BS) may prevent the flames or gas (E) from flowing into the accommodating space S. Thereby, the battery module **50** of the present disclosure may delay as much as possible the occurrence of thermal runaway in one battery cell **10**, which can cause a short circuit in the adjacent battery cell **10**.

[0091] As set forth above, according to an embodiment of the present disclosure, a battery cell may discharge flames, gas, or the like, within an accommodating space to the outside of a case when an event has occurred, but may prevent flames, gases, or the like, from the outside of the case, from flowing into the accommodating space.

[0092] According to an embodiment of the present disclosure, a battery module may discharge flames, gas, or the like, within an accommodating space to the outside of a case, when an event has occurred, but may prevent flames, gas, or the like, generated within an adjacent a battery cell from flowing into the battery module.

[0093] While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed to have a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

[0094] The above-described description is merely an example of applying the principles of the present disclosure, and other configurations may be further included without departing from the scope of the present disclosure.

Claims

1. A battery cell, comprising: a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; a vent provided in the case or the cap plate; and a venting cover disposed to cover the vent, and provided to be opened by pressure acting from the accommodating space to the outside of the case, wherein the venting cover includes a venting sheet provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent and a second surface opposite to the first surface; and a venting cap having a support portion supporting the first surface of the venting sheet.
2. The battery cell of claim 1, wherein the venting sheet includes a venting line formed on at least one of the first surface or the second surface.
3. The battery cell of claim 2, wherein the venting sheet is disposed to face the vent in a first

direction, and the support portion is disposed to correspond to the venting line, and at least a portion of the support portion overlaps the venting line in the first direction.

4. The battery cell of claim 2, wherein the venting line is formed by being partially recessed from the second surface.

5. The battery cell of claim 1, wherein the vent is provided to be opened by a predetermined amount of pressure and to communicate the accommodating space and the outside of the case.

6. The battery cell of claim 5, wherein the vent includes a notch groove formed by being at least partially recessed in a thickness direction.

7. The battery cell of claim 1, wherein the venting cover is provided to be opened when pressure within the accommodating space becomes greater than pressure outside the case by a predetermined value, to be opened by force toward the outside of the case from the accommodating space.

8. The battery cell of claim 1, wherein the venting cover further comprises a venting space formed between the venting sheet and the vent, the vent is provided to be opened by a pressure difference between the accommodating space and the venting space, and the venting sheet is provided to be opened toward the outside of the case when the vent is opened.

9. The battery cell of claim 1, wherein the vent includes a notch recessed at least partially in the first direction.

10. The battery cell of claim 1, wherein the support portion is disposed between the venting sheet and the vent.

11. The battery cell of claim 1, further comprising a terminal portion disposed on the cap plate and electrically connected with the electrode assembly, wherein the terminal portion includes: a cathode terminal electrically connected with a cathode tab of the electrode assembly; and an anode terminal electrically connected with an anode tab of the electrode assembly.

12. The battery cell of claim 11, wherein the venting cover is disposed between the cathode terminal and the anode terminal.

13. The battery cell of **12**, wherein the vent is disposed between the cathode terminal and the anode terminal.

14. The battery cell of claim 1, wherein the venting lines is plurality, and the plurality of venting lines are arranged in a second direction perpendicular to the first direction.

15. The battery cell of claim 1, wherein the support portion is provided in plurality, and the plurality of support portions are arranged in the second direction.

16. A battery cell, comprising: a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; and a venting cover provided in the cap plate or the case, and provided to be opened from the accommodating space in a direction toward the outside of the case, wherein the venting cover includes: a venting sheet provided to be opened by a predetermined amount of pressure, and including a venting line formed on at least one of a first surface facing the accommodating space and a second surface opposite to the first surface in a first direction; and a venting cap having a support portion supporting the first surface of the venting sheet, wherein at least a portion of the support portion is disposed to overlap the venting line in the first direction.

17. A battery module, comprising: a plurality of battery cells; and a module housing accommodating the plurality of battery cells, wherein at least one of the plurality of battery cells includes a case accommodating an electrode assembly in an accommodating space; a cap plate coupled to at least one side of the case; a vent provided in the case or the cap plate; and a venting cover disposed to cover the vent, and provided to be opened by pressure acting from the accommodating space to the outside of the case, wherein the venting cover includes: a venting sheet provided to be opened by a predetermined amount of pressure, and including a first surface facing the vent and a second surface opposite to the first surface; and a venting cap having a support portion supporting the first surface of the venting sheet.

